



EDU TERIA

72nd BPSC

Prelims Online Batch

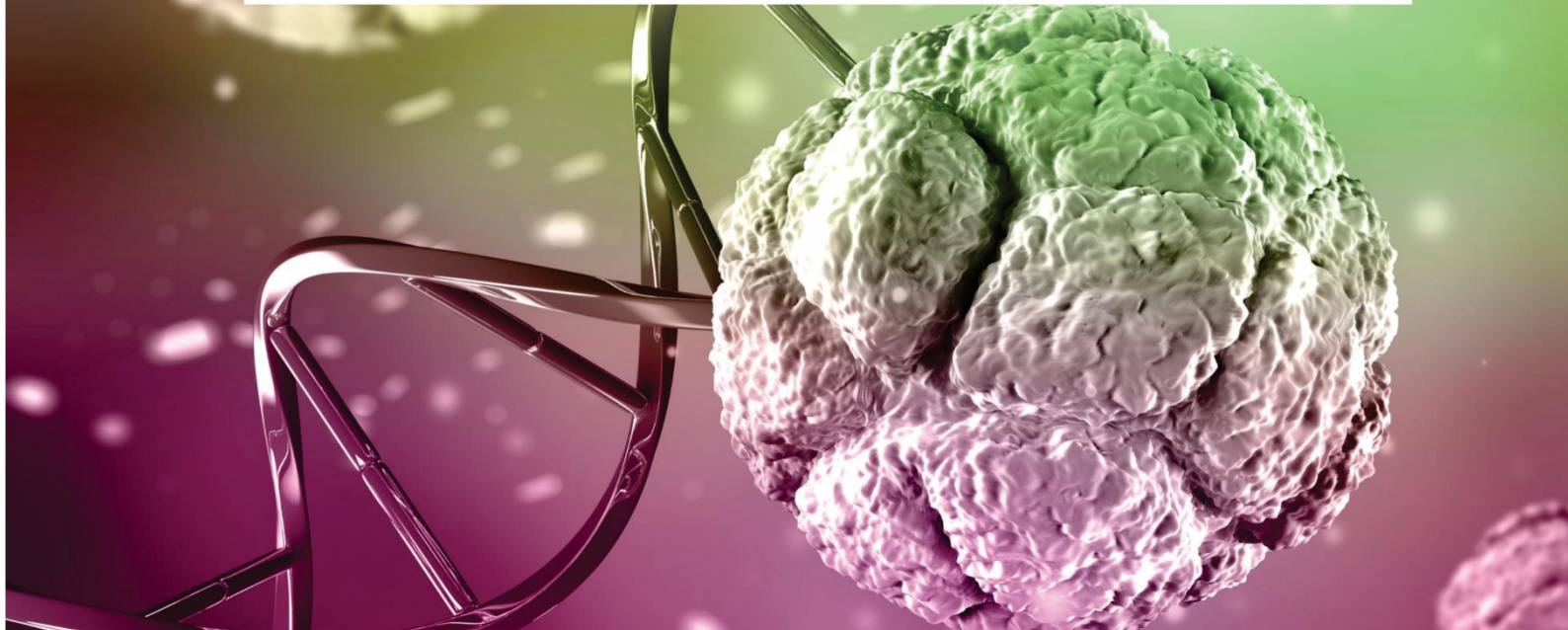
BIOLOGY

Topic wise
Notes

LECTURE

22

Nervous System - 01



Nervous System - 01

Nervous System

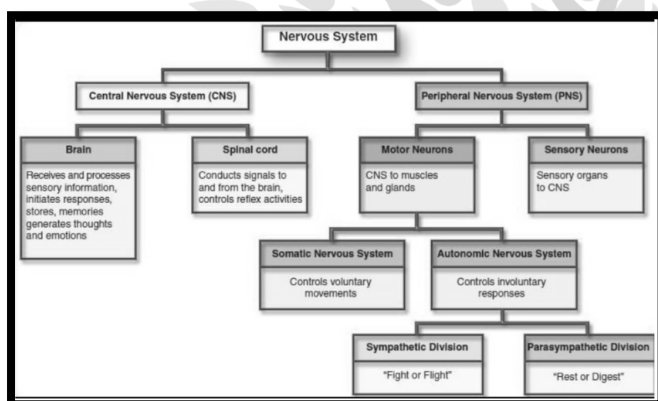
The nervous system controls and coordinates all activities of the body. It helps the body respond to internal and external stimuli. Information is transmitted in the form of electrical impulses.

Divisions of Nervous System

1. Central Nervous System (CNS)

- Brain
- Spinal cord

Functions: Processing, integrating, and responding to information.



2. Peripheral Nervous System (PNS)

- Cranial nerves (12 pairs)
- Spinal nerves (31 pairs)

Functions: Connects CNS to body parts.

PNS is further divided into:

A) Somatic Nervous System

- Voluntary actions
- Controls of skeletal muscles

Peripheral Nervous System			
Autonomic nervous system		Somatic nervous system	
Parasympathetic division	Sympathetic division	Sensory system (afferent)	Motor system (efferent)

B) Autonomic Nervous System (ANS)

- Involuntary activities
- Controls cardiac muscle, smooth muscle, glands

- ANS has two parts:

Part	Function
Sympathetic Nervous System	Flight-or-flight; increases heart rate, breathing rate
Parasympathetic Nervous System	Rest and digest, slow heart rate, promotes digestion

Brain

The brain is the main control center of the nervous system. It processes information, controls body activities, and enables thinking, memory, emotions, and coordination.

1. The brain is the central information processing organ of our body and acts as the 'Command and control system'.
2. It controls the voluntary movements, balance of the body, functioning of vital involuntary organs (e.g. lungs, heart, kidney etc.), thermoregulation, hunger and thirst, circadian (24 hours) rhythms of our body, activities of several endocrine glands and human behaviour.
3. It is also site for processing of vision, hearing, speech, memory, intelligence, emotions and thoughts.
4. It is situated in cranial box which is made up of 1 frontal bone, 2 parietal bone, 2 temporal bone, 1 sphenoid, 1 ethmoid 1 occipital bone. The weight of brain of an adult male is 1400 gm and of female is 1250 gm.

Structure and Functions of Brain:

Brain is protected by cranium (skull) and meninges (three membranes):

1. Dura mater

- This is the first and the outermost membrane which is thick, strong and elastic layer.
- At several places it forms cranial venous sinuses containing blood



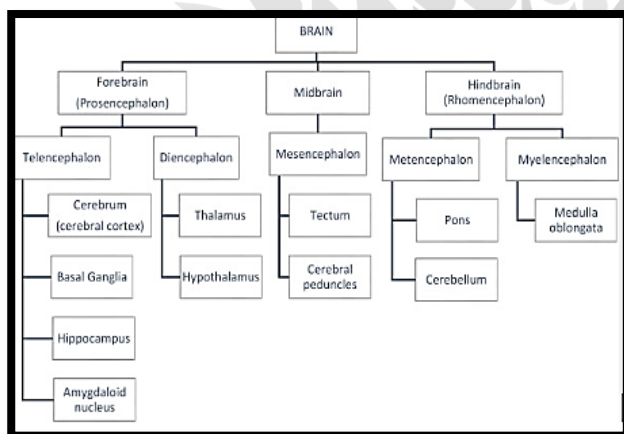


2. Arachnoid mater

- * It is middle and delicate layer and found only in mammals. (Mammalian character)
- * At several places it forms villi like foldings to absorb CSF called arachnoid villi.
- * Space between duramater & arachnoid is called subdural space which is filled with serous fluid.

3. Pia mater

- * Inner most, thin and transparent membrane, which is firmly attached to the brain.
- * At some places it merges deeply into sulci of brain to form telachoroidea.
- * Space between arachnoid & piamater is called subarachnoid space, which is filled with CSF.



Brain regions:

1. Forebrain

- The fore brain consists of Cerebrum, Diencephalon (containing epithalamus, thalamus, hypothalamus) and olfactory lobe. Cerebrum forms the major part of the brain which is most developed in human. Cerebrum consists of two cerebral hemispheres, on the dorsal surface a longitudinal groove is present between two cerebral hemispheres called as median fissure. Both the cerebral hemispheres partially connected with each other by curved white thick nerve fibre called corpus callosum (Largest commissure of brain) (Mammalian character).
- * Each cerebral hemisphere is divided into 4 lobes—Anterior, Middle, Posterior and Lateral.

- * Anterior lobe is also called frontal lobe (largest lobe) while middle lobe is called as parietal lobe. Frontal lobe is separated by central sulcus or rolandic sulcus from parietal lobe.

Function of cerebrum : It is the most important part of brain because it controls and regulates different part of brain. This is the centre of conscious senses, will power. Voluntary movements, knowledge, etc.

- * Different sense organs send impulse here and in this part of brain, analysis and coordination of impulses is done then messages are transferred to organs.

Diencephalon

- * It is small chamber like posterior part of fore brain which is covered by cerebrum. It consists of 3 parts :

(i) Epithalamus (ii) Thalamus (iii) Hypothalamus

(i) Epithalamus : It forms the roof of diencephalon. Pineal body (Epiphysis cerebri) is found on epithalamus & control sexual maturity.

(ii) Thalamus : It forms upper lateral wall of Diencephalon. It forms 80% of diencephalon. It acts as a relay centre. It receives all sensory inputs from all part of body & these impulses are sent to the cerebral cortex. Cerebrum wraps around the thalamus. It is a major coordinating centre for sensory & motor signalling.

(iii) Hypothalamus : It forms the lower or ventral part of diencephalon. It lies at the base part thalamus.

- * The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking (Hunger and thirst).
- * It also contains several groups of neurosecretory cells, which secrete hormone called hypothalamic hormone.
- * A cross like structure is found on anterior surface of hypothalamus called as optic chiasma, through Infundibulum pituitary body is attached to middle part of hypothalamus. Corpus mammillare or corpus albicans or mammillary body is found on the posterior part of hypothalamus. (Mammalian character)



**Hypothalamus controls :-**

- (1) Thermoregulation
- (2) Behaviour and emotion
- (3) Endocrine control
- (4) Biological clock system
- (5) ANS control.

Mid Brain

It is a small part of brain. The midbrain is located between diencephalon of the fore brain and pons of the hind brain. A canal called Cerebral aqueduct/Aqueduct of Sylvius passes through the midbrain.

- * Anterior part of midbrain contains two longitudinal myelinated nerve fibre called cerebral peduncles or crus cerebri or crura cerebri.
- * On posterior part of mid brain four spherical projections are found called colliculus or optic lobe. Four colliculus are collectively called as corpora quadrigemina. (2 upper & 2 lower) (Mammalian character)
- * Mid brain and hind brain (except cerebellum) form the brain stem. Function :
- * The mid brain receives and integrates visual, tactile and auditory inputs.

Hindbrain

- * The hind brain comprises pons, cerebellum and medulla (also called the medulla oblongata).
- * PONS
- * It is a small spherical projection. Which is situated below the midbrain and upper side of medulla oblongata. It consists of many transverse and longitudinal nerve fibre. Transverse nerve fibre connect with cerebellum while longitudinal nerve fibre connect cerebrum to medulla oblongata.

Function :- It regulates the breathing reaction through pneumotaxic centre.

CEREBELLUM

- * It is made up of 3 lobe (2 lateral lobe and 1 vermis). Both lateral lobes become enlarged and spherical in shape,

so lateral lobe of cerebellum are also called as cerebellar hemisphere. Cerebellum has very convoluted surface in order to provide the additional space of for many more neurons. Posterior part of hind brain is tubular and cylindrical in shape. Medulla of brain is connected to spinal cord.

- * Mid brain, pons and medulla are situated in one axis and it is called as Brain stem.

Functions : It is the most important part of brain which controls all the involuntary activities of the body e.g. cardiovascular reflex, respiration, metabolism, gastric secretion etc. As well as this act as conduction path for all impulses between spinal cord and remaining portions of brain.

- * It is also concerned with cranial reflex action like sneezing reflex, salivation reflex, coughing reflex, swallowing reflex, vomiting reflex, yawning reflex.

Brain